# PES UNIVERSITY

## DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

ACADEMIC YEAR 2025-2026 | FIFTH SEMESTER B.TECH

**SOFTWARE TEST PLAN (STP Specification)**

# PROJECT: CAREERBRIDGE

*Online Internship & Placement Preparation Portal*

### SUBMITTED BY: TEAM 4


| Student Name | USN | Role Assigned | Deliverable |
|---|---|---|---|
| Deepika K | PES2UG24CS148 | Team Lead & Architect | Architecture, ADRs, Component Diagram |
| Gunavathi M I | PES2UG24CS172 | Frontend & UI/UX Lead | Design Overview, UX Design, Sequence Diagrams |
| Darshan R Mali | PES2UG24CS144 | Backend & Database Architect | API Design, Data Stores, Security Architecture |
| Dhanush Rao | PES2UG24CS153 | Full-Stack & QA/DevOps Lead | Test Strategy, Metrics, CI/CD Pipeline |


Version: 1.0 (Draft for Review) | Date: 15-09-2026 | Status: Draft

*Repository: <to be added by Team 4>*

**Software Test Plan**

Version: 1.0
Authors: Team 4 (CareerBridge)
Date: 15-09-2026
Status: Draft for Faculty Review

## Revision History

| Version | Date | Author | Change Summary |
|---|---|---|---|
| 0.1 | 12-09-2026 | Team 4 | Initial test-item and scope draft |
| 1.0 | 15-09-2026 | Team 4 | Complete STP with test strategy, 8 FR/6 NFR mappings, security validation, RTM, and metrics framework |

# 1. Introduction

Purpose: This document defines the Software Test Plan (STP) for CareerBridge v1.0, an online internship and placement preparation portal. It establishes the objectives, scope, strategy, resources, schedule, and responsibilities for all testing activities, ensuring that the delivered system satisfies the functional and non-functional requirements specified in the Software Requirements Specification (SRS v1.0).

Scope: Testing covers all CareerBridge features including student authentication, internship browsing and filtering, application submission and tracking, aptitude test attempts with adaptive difficulty, progress analytics, report export, and the CI/CD pipeline. Excluded from testing scope: third-party job-portal integration internals, browser rendering engine internals, and operating-system-level network behavior.

**References:**

- CareerBridge SRS v1.0
- CareerBridge SAD v1.0 (companion document)
- Requirements Traceability Matrix (SRS Section 8)
- Database Schema: database/schema.js (Mongoose)
- CI/CD Workflow: .github/workflows/ci.yml
- IEEE 829: Standard for Software and System Test Documentation

## Definitions

| Term | Definition |
|---|---|
| STP | Software Test Plan |
| SRS | Software Requirements Specification |
| SAD | Software Architecture and Design Specification |
| RTM | Requirements Traceability Matrix |
| FR | Functional Requirement |
| NFR | Non-Functional Requirement |
| UAT | User Acceptance Testing |
| CI/CD | Continuous Integration / Continuous Deployment |
| XSS | Cross-Site Scripting |
| JWT | JSON Web Token |

# 2. Test Items

The following CareerBridge modules constitute the test items for this plan:

| Module | Source Path | Description |
|---|---|---|
| Authentication Module | src/context/AuthContext.jsx | Student registration, login, session management (JWT) |
| Internship Listings Module | src/components/listings/InternshipListings.jsx | Browse, search, and filter internship postings |
| Application Module | src/components/applications/ApplicationTracker.jsx | Application submission, status tracking |
| Aptitude Test Module | src/components/tests/AptitudeTestPanel.jsx + analytics_service/test_engine.py | Timed test delivery, scoring, adaptive difficulty |
| Progress Analytics Module | src/components/dashboard/ProgressDashboard.jsx + analytics_service/analytics.py | Readiness score, trend visualization |
| Notification Module | src/components/notifications/NotificationCenter.jsx | In-app and email/SMS notifications |
| Report Export Module | analytics_service/reports.py | PDF/CSV report generation |

# 3. Features to be Tested

## 3.1 Functional Requirements

| Req ID | Feature Description | Test Priority |
|---|---|---|
| CB-F-001 | Student registration and login (JWT-based) | High |
| CB-F-002 | Browse and filter internship listings by domain/location/stipend | High |
| CB-F-003 | Apply to an internship with resume upload | High |
| CB-F-004 | Track application status (applied/shortlisted/offered/rejected) | High |
| CB-F-005 | Attempt a timed aptitude test with adaptive difficulty | High |
| CB-F-006 | View readiness score and topic-level progress trends | Medium |
| CB-F-007 | Export progress/test report as PDF or CSV | Medium |
| CB-F-008 | Receive email/SMS notification on application status change | Medium |

## 3.2 Non-Functional Requirements

| Req ID | Feature Description | Test Priority |
|---|---|---|
| CB-NF-001 | Internship search results returned within 2 seconds | High |
| CB-NF-002 | System availability target of 99.5% during placement season | Medium |
| CB-NF-003 | Cross-browser compatibility (Chrome, Firefox, Edge) | High |
| CB-NF-004 | Data encryption in transit via TLS 1.2+ | High |
| CB-NF-005 | Support 500+ concurrent internship listing records | Medium |
| CB-NF-006 | Graceful degradation when analytics microservice is unreachable | High |

# 4. Features Not to be Tested

- Third-party job-portal integration API internals (assumed reliable per vendor SLA)
- Browser rendering engine CSS layout calculations
- Email/SMS provider (Nodemailer/Twilio) delivery internals
- Operating system-level network stack behavior
- Node.js runtime internals and V8 engine optimizations
- npm/pip package registry availability during dependency installation

# 5. Test Approach / Strategy

## 5.1 Test Levels

| Level | Description | Tools | Scope |
|---|---|---|---|
| Unit Testing | Isolated testing of individual functions and utility modules | Jest, PyTest | Scoring logic, readiness calculations, validation |
| Integration Testing | Testing interactions between components and the data/service layer | Jest, Postman | AuthContext-Express API, Express-Flask analytics call |
| System Testing | End-to-end testing of complete user workflows | Manual browser testing | Registration-to-application flow, test-attempt-to-dashboard flow |
| Acceptance Testing (UAT) | Validation against SRS acceptance criteria by evaluators | Manual demonstration | All 8 FRs verified against acceptance criteria |

## 5.2 Test Types

| Type | Purpose | Applicable Requirements |
|---|---|---|
| Functional Testing | Verify correct behavior of all system features | CB-F-001 through CB-F-008 |
| Regression Testing | Ensure new changes do not break existing functionality | All requirements after code modifications |
| Performance Testing | Validate response time and throughput targets | CB-NF-001, CB-NF-005 |
| Usability Testing | Assess UI clarity, navigation flow, and accessibility | CB-NF-003, UX requirements |
| Security Testing | Validate authentication, authorization, and data protection | CB-NF-004, security validation items |

## 5.3 Entry and Exit Criteria

**Entry Criteria:**

- Stable build delivered (npm run build and flask run complete without errors)
- All unit tests passing (0 failures in Jest and PyTest)
- Test environment ready (Node.js 20+, Python 3.x, browser available)
- Test data available (seed data script populated)
- SRS and SAD documents reviewed and approved

**Exit Criteria:**

- 100% of planned test cases executed
- 0 critical or high-severity defects remain open
- All 8 functional requirements verified against acceptance criteria
- All 6 non-functional requirements measured and within threshold
- Requirements Traceability Matrix shows complete coverage
- Test Summary Report reviewed and approved

### 5.4 Security Validation

| Security Test | Validation Method | Pass Criteria |
|---|---|---|
| XSS Input Sanitization | Inject script tags and event handlers into all text fields (cover note, profile bio) | No script execution; input rendered as plain text via React JSX auto-escaping |
| JWT Session Handling | Examine browser storage for tokens after login; test expired-token rejection | Tokens stored securely; expired tokens rejected on subsequent API calls |
| Authorization / Ownership Checks | Attempt to fetch another student's application or resume via direct API calls | 403 Forbidden response; zero unauthorized records returned |
| TLS Enforcement | Inspect network traffic using browser DevTools during API communication | All requests use HTTPS; no plaintext HTTP observed |
| File Upload Safety | Attempt to upload non-resume file types and oversized files | Rejected with a clear validation error; only PDF/DOCX under size limit accepted |
| Input Validation Fuzzing | Submit malformed data (empty strings, excessively long strings, special characters) to all forms | Application handles gracefully with validation errors; no crashes |

## 6. Test Environment

### 6.1 Hardware

| Component | Specification |
|---|---|
| Development Machine | Windows 10/11 or macOS, 8GB+ RAM, modern x86-64/ARM processor |
| Display | Minimum 1024x768 resolution for responsive layout testing |
| Network | Broadband connection for MongoDB Atlas testing; disconnection simulation for offline-degradation testing |

### 6.2 Software

| Component | Version | Purpose |
|---|---|---|
| Node.js | 20.0.0+ | JavaScript runtime for the Express API and build tooling |
| Python | 3.10+ | Runtime for the Flask analytics microservice |
| npm / pip | Latest | Package managers for JS and Python dependencies |
| Vite | 5.x | Development server and production bundler |
| Jest / PyTest | Latest | Unit and integration test frameworks |
| Google Chrome | Latest | Primary testing browser |
| Mozilla Firefox | Latest | Cross-browser compatibility testing |

| Component | Version | Purpose |
|---|---|---|
| Microsoft Edge | Latest | Cross-browser compatibility testing |
| Postman | Latest | API testing for Express and Flask endpoints |
| GitHub Actions | Latest | CI/CD pipeline execution |

### 6.3 Test Data

- Seed script providing 4 sample student profiles (one per team member)
- 20 sample internship listings across multiple domains and locations
- Sample aptitude question bank covering 5 topics at 3 difficulty levels
- Pre-populated test attempt history to validate readiness-score computation

## 7. Test Schedule

| Milestone | Planned Date | Status |
|---|---|---|
| Test Plan finalization | 01-10-2026 | Planned |
| Test case design and review | 03-10-2026 | Planned |
| Test environment setup and validation | 03-10-2026 | Planned |
| Unit test execution (Jest + PyTest) | 05-10-2026 | Planned |
| Integration test execution | 08-10-2026 | Planned |
| System test execution (end-to-end workflows) | 12-10-2026 | Planned |
| Security validation execution | 15-10-2026 | Planned |
| Performance test execution | 17-10-2026 | Planned |
| User Acceptance Testing (UAT) | 20-10-2026 | Planned |
| Test Summary Report generation | 23-10-2026 | Planned |

## 8. Test Deliverables

| Deliverable | Description |
|---|---|
| Software Test Plan (this document) | Defines scope, strategy, schedule, and responsibilities for all testing activities |
| Test Cases | Detailed test case specifications mapping to SRS requirement IDs |
| Automated Test Scripts | Jest test files (src/tests/*.test.js) and PyTest files (analytics_service/tests/*.py) |
| Test Data | Seed data configuration scripts |
| CI/CD Test Execution Logs | GitHub Actions workflow run logs capturing build, test, and lint results |
| Test Summary Report | Final summary of test execution results, defect statistics, and coverage metrics |

# 9. Roles and Responsibilities

| Role | Name | Responsibility |
|---|---|---|
| QA Lead | Deepika K (PES2UG24CS148) | Prepare test plan, coordinate test execution, compile test summary report |
| Test Engineer (Frontend) | Gunavathi M I (PES2UG24CS172) | Design and execute UI/UX test cases, usability testing, cross-browser validation |
| Test Engineer (Backend) | Darshan R Mali (PES2UG24CS144) | Design and execute API/data-service test cases, database integration testing, security testing |
| Test Architect | Dhanush Rao (PES2UG24CS153) | Define test strategy, review test cases, approve test results, CI/CD pipeline oversight |

# 10. Risks and Mitigation

| Risk | Probability | Impact | Mitigation |
|---|---|---|---|
| MongoDB Atlas sandbox unavailability during integration testing | Medium | High | Maintain a local MongoDB instance as fallback; schedule cloud tests during off-peak hours |
| Cross-browser rendering inconsistencies affecting test reliability | Medium | Medium | Standardize testing on Chrome as primary browser; document browser-specific deviations |
| Python analytics microservice downtime blocking dashboard tests | Low | High | Mock the analytics service response in integration tests; test the graceful-degradation path separately |
| Insufficient test data for edge-case coverage | Low | Medium | Extend seed data to include boundary conditions (empty listings, max file size uploads, 100+ applications) |
| CI/CD pipeline timeout on GitHub Actions free tier | Low | Low | Optimize test execution time; split long-running jobs in parallel |
| Dependency version conflicts after install | Low | Medium | Lock dependency versions (package-lock.json, requirements.txt); use clean installs in CI |

# 11. Assumptions and Dependencies

**Assumptions:**

- Node.js 20+ and Python 3.10+ are available on all testing machines
- MongoDB Atlas sandbox will be accessible for cloud integration testing
- Seed data provides sufficient coverage for functional testing
- Browser DevTools are available for network simulation and performance profiling
- GitHub Actions runners are available for CI/CD pipeline execution

**Dependencies:**

- SRS v1.0 document must be finalized before test case design
- SAD v1.0 document must be finalized before integration test design
- Application codebase must compile and pass unit tests before system testing
- MongoDB Atlas project must be provisioned for cloud database integration testing

- GitHub repository must be configured with a CI/CD workflow for automated test execution

# 12. Suspension and Resumption Criteria

**Suspend testing if:**

- Production build fails (npm run build or flask run returns a non-zero exit code)
- More than 30% of unit tests fail, indicating systemic instability
- Test environment is unavailable for more than 4 hours
- A critical defect is discovered that blocks further test execution
- MongoDB Atlas experiences a documented outage affecting all regions

**Resume testing when:**

- Build failures are resolved and verified by a clean build
- Blocking defects are fixed, verified, and merged to the test branch
- Test environment is restored and validated
- MongoDB Atlas service status returns to operational

# 13. Test Case Management and Traceability

The Requirements Traceability Matrix ensures bidirectional mapping between SRS requirements and test cases. Each requirement ID is traced to one or more test cases with defined pass/fail criteria:

## 13.1 Functional Requirements Traceability

| Req ID | Requirement | Test Case ID | Expected Result |
|---|---|---|---|
| CB-F-001 | Student authentication | TC-AUTH-01 | User profile displayed in Navbar; JWT session persisted |
| CB-F-002 | Browse/filter internships | TC-LIST-01 | Results match applied filters within 2 seconds |
| CB-F-003 | Apply to internship | TC-APP-01 | Application appears in tracker with status 'Applied' |
| CB-F-004 | Application status tracking | TC-APP-02 | Status updates reflected after recruiter action |
| CB-F-005 | Aptitude test attempt | TC-TEST-01 | Score and breakdown returned; next difficulty adjusted |
| CB-F-006 | Progress/readiness dashboard | TC-DASH-01 | Readiness score and trend chart render correctly |
| CB-F-007 | Report export | TC-REP-01 | PDF/CSV downloads with correct data |
| CB-F-008 | Notification on status change | TC-NOTIF-01 | Email/SMS received within expected time window |

## 13.2 Non-Functional Requirements Traceability

| Req ID | Requirement | Test Case ID | Pass Criteria |
|---|---|---|---|
| CB-NF-001 | Search response time | TC-PERF-01 | Listing search returns in under 2 seconds |
| CB-NF-002 | 99.5% availability | TC-AVAIL-01 | No unplanned downtime observed during test window |
| CB-NF-003 | Cross-browser support | TC-COMPAT-01 | Core workflows function identically on Chrome, Firefox, Edge |
| CB-NF-004 | TLS encryption | TC-SEC-01 | All API requests use HTTPS with TLS 1.2+ |
| CB-NF-005 | 500+ listing capacity | TC-LOAD-01 | UI remains responsive with 500 seeded listings |

| Req ID | Requirement | Test Case ID | Pass Criteria |
|---|---|---|---|
| CB-NF-006 | Graceful analytics degradation | TC-RECOV-01 | Cached readiness score shown when analytics service is down |

# 14. Test Metrics and Reporting

## 14.1 Metrics Collected

| Metric | Definition | Target |
|---|---|---|
| Test Case Execution Rate | Percentage of planned test cases executed | 100% |
| Test Pass Rate | Percentage of executed test cases that passed | 95% or higher |
| Defect Density | Number of defects per module | Less than 2 per module |
| Defect Aging | Average time from defect discovery to resolution | Less than 24 hours |
| Requirement Coverage | Percentage of SRS requirements with at least one mapped test case | 100% |
| Code Coverage (Unit Tests) | Percentage of utility/service code exercised by Jest/PyTest | 80% or higher |
| CI/CD Pass Rate | Percentage of GitHub Actions workflow runs that pass | 95% or higher |

## 14.2 Reporting

| Report | Frequency | Audience |
|---|---|---|
| Daily Test Execution Status | During active test execution phase | Team Lead, QA Lead |
| Defect Triage Report | Per defect discovery | Developer assigned to fix |
| CI/CD Pipeline Status | Per push/PR (automated) | All team members |
| Final Test Summary Report | End of testing cycle | Course Evaluators, Team Lead |

## 14.3 Test Summary Report Contents

The final Test Summary Report will include:

- Total test cases planned, executed, passed, failed, and blocked
- Defect summary (total found, fixed, deferred, by severity)
- Requirement coverage matrix showing 100% traceability
- CI/CD pipeline execution history with pass/fail statistics
- Recommendations for future test cycle improvements

# 15. Approvals

| Role | Name | Signature / Date |
|---|---|---|
| QA Lead | Deepika K (PES2UG24CS148) | |
| Test Architect | Dhanush Rao (PES2UG24CS153) | |
| Course Coordinator | Software Engineering Faculty | |